

Shivam Kumar

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Education

Year	Degree	Institute	GPA/%
2024-2028	B.Tech, Computer Science	Delhi Technological University, New Delhi	8.73
2023	12th CBSE	New Modern Public School, Bihar	88%
2021	10th CBSE	Vidya Vihar Residential School, Bihar	94%

Technical Skills

Languages:	Python, C/C++, JavaScript, TypeScript, SQL
Cloud & DevOps:	AWS (Lambda, Amplify), Serverless Architecture, Docker, Git, GitHub, Linux
Backend:	Node.js, Next.js, Flask, REST APIs, Microservices
Databases:	PostgreSQL (SQL), Redis, MongoDB, Firebase (NoSQL)
Core CS & AI:	DSA, OOP, Operating Systems, DBMS, Computer Networks, CLIP, FAISS

Work Experience

AI Generalist, Young Founders School Jul 2025 – Present

- Migrated the production backend of **Workskills** (Next.js, **8,000+ monthly active users**) from an n8n workflow engine to **serverless AWS Lambda**, deploying the full stack on **AWS Amplify** to improve scalability
- Architected **Bootcamp AI** from the ground up as a full-stack AI learning product on Next.js with a **cloud-native AWS Lambda and Amplify backend**, owning the system end to end from initial design through deployment
- Engineered a **Google Apps Script** system auto-assigning students across **11 facilitators** via a weighted workload algorithm, and authored a **34-point automation roadmap** adopted by Programs & Courses leadership

Software Developer, FocusDesk Jun 2026 – Present

- Architected backend microservices (**Node.js, PostgreSQL, Redis, REST APIs**) powering authentication, session orchestration, and real-time workflows for a scalable, concurrent user-management system in production
- Reduced **API latency 30%** (~200ms → ~140ms) via Redis caching and normalized PostgreSQL schemas

Full Stack Developer, TripXpay May 2025 – Sep 2025

- Shipped platform features (**Next.js, Firebase**) for E-KYC workflows and credit transactions, building 5+ backend services and Firebase schemas for efficient querying and reliable, secure data interactions across user and admin flows
- Migrated React to **Next.js with SSR** to improve page performance, and resolved production-level issues across core user and admin workflows by reproducing failures, tracing them to root cause, and shipping targeted fixes

Selected Technical Projects

Deep Packet Inspection System | [GitHub](#) C++, Multithreading, Networking

- Architected a **multithreaded DPI engine** parsing L2-L7 protocols from PCAP traffic at **8K+ pkts/sec**
- Restructured the concurrent pipeline to lift **throughput 35%** and cut latency ~18ms, using profiling and lock-contention analysis to isolate bottlenecks and rebalance work across worker threads in the packet-processing path
- Implemented TLS SNI extraction and rule-based classification for low-overhead traffic analysis

Pic2Pick – Semantic Image Similarity System | [GitHub](#) | [Ⓛive](#) Python, CLIP, FAISS

- Engineered a **CLIP embedding pipeline** with **FAISS ANN search** for scalable semantic image retrieval
- Reduced retrieval **latency 30%** (~480ms → ~340ms) via index tuning and preprocessing, benchmarking trade-offs across embedding models, index types, and query latency to guide retrieval design while preserving accuracy

Ethicly – AI Fairness Auditing Platform | [GitHub](#) | [Ⓛive](#) Python, Flask, Gemini API, PostgreSQL

- Built an **algorithmic fairness auditing platform** computing statistical metrics (demographic parity, approval gap, fairness score) over user-supplied tabular datasets, with audit runs and computed metrics persisted in PostgreSQL
- Designed AI-assisted audit reporting and bias visualizations translating quantitative metrics into readable insights

Achievements

- Winner, National Project Competition** organized by **ISRO** (2023) | [Certificate](#)
- Dell Aspire Scholar**: merit-based, MSDF-backed scholarship with mentorship and career support
- Core member, **EHAX** – DTU's Cybersecurity Society; co-organized **EHAX CTF**, a capture-the-flag event with 2,000+ participants, which earned the society **India Rank 7** and **World Rank 33** in CTFtime's global rankings
- Talerang'26 25 Under 25**: recognized for innovation across the ISRO-winning project and [Ecoshala](#)